

■ BusinessNext

# AWS Cost Estimator

Pricing & Infrastructure Report

PNB

Mode: On-Premise — Client Hosted

Date: 09 April 2026

Inflation: 4% / yr

AWS Region: us-east-1

# Executive Summary

High-level cost overview and infrastructure snapshot

|                     |                    |              |                 |
|---------------------|--------------------|--------------|-----------------|
| 9                   | 96                 | 172          | 224             |
| Worker Nodes        | vCPUs              | RAM (GB)     | DB RAM (GB)     |
| Total compute nodes | Total virtual CPUs | Total memory | Database memory |

## Project Details

|                       |                            |
|-----------------------|----------------------------|
| Customer / Bank       | PNB                        |
| Client Mode           | On-Premise — Client Hosted |
| Database Type         | SQL Server                 |
| Total Named Users     | 15,500                     |
| Concurrent Users      | 4,650 (30% of named)       |
| Mobile Users          | 4,050                      |
| Worker Nodes          | 9                          |
| Total vCPUs (worker)  | 96                         |
| Total RAM GB (worker) | 172 GB                     |
| Data Size             | 2100 GB                    |
| Estimate Date         | 09 April 2026              |

# Node Distribution

AI-assisted infrastructure architecture

|                       |                       |
|-----------------------|-----------------------|
| Total Worker Nodes    | 9                     |
| Total DB Nodes        | 13                    |
| Fixed / Managed Roles | 7                     |
| Confidence            | High                  |
| Distribution Method   | Groq LLM + Rule-based |

| Category                    | Role / Service                                    | Nodes | Instance Fam.         | vCPU/node | RAM GB/node | Model         |
|-----------------------------|---------------------------------------------------|-------|-----------------------|-----------|-------------|---------------|
| Kubernetes Cluster          | Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)     | 6     | Compute Intensive     | 16        | 32          | Reserved 3 Yr |
| Kubernetes Cluster          | Worker Nodes Linux/RHEL for Graphana & Prometheus | 1     | Compute Intensive     | 8         | 16          | Reserved 3 Yr |
| Kubernetes Cluster          | Worker Nodes Linux/RHEL for EFK - optional        | 2     | Compute Intensive     | 8         | 16          | Reserved 3 Yr |
| PGSQL Database              | Linux Nodes – Database server (Primary)           | 2     | Memory Intensive      | 32        | 128         | None          |
| PGSQL Database              | Linux Nodes – Cluster (etcd+haproxy, pgbackrest)  | 4     | Memory Intensive      | 2         | 8           | None          |
| PGSQL Database              | Storage: SAN (Primary DB)                         | 2     | 10K IOPS              | —         | —           | None          |
| S3                          | S3 Storage + Replication                          | 1     | Cloud Storage Service | —         | —           | On Demand     |
| PGSQL Database – Reporting  | Linux Nodes – Database server (Reporting)         | 1     | Memory Intensive      | 32        | 128         | None          |
| PGSQL Database – Reporting  | Storage: SAN (Reporting DB)                       | 1     | 10K IOPS              | —         | —           | None          |
| Security & Compliance       | WAF + AWS Shield Advanced                         | 1     | Managed Service       | —         | —           | On Demand     |
| Security & Compliance       | CloudTrail + Config Audit Logs                    | 1     | Managed Service       | —         | —           | On Demand     |
| Kubernetes Cluster          | Cloud Managed K8s                                 | 1     | K8s Service           | —         | —           | None          |
| Caching & Session           | Elastichache Service                              | 2     | Memory Optimized      | 4         | 16          | Reserved 3 Yr |
| Infrastructure & Monitoring | Load Balancer (Internal + External)               | 2     | ALB                   | —         | —           | On Demand     |
| Infrastructure & Monitoring | Bastion Host                                      | 1     | General Purpose       | 2         | 4           | Reserved 3 Yr |
| Infrastructure & Monitoring | NAT Gateway                                       | 2     | NAT                   | —         | —           | On Demand     |
| Infrastructure & Monitoring | Image Registry (ECR)                              | 1     | Managed Service       | —         | —           | On Demand     |
| Infrastructure & Monitoring | Back Up - DB & Infra Logs                         | 1     | Cloud Storage Service | —         | —           | On Demand     |

# Environment Pricing

Pre-Prod / SIT / UAT and DR environment costs

## Environments: SIT, UAT

|                        |                                        |
|------------------------|----------------------------------------|
| Environments Selected  | 2 — SIT, UAT                           |
| Base Cost per Env / mo | \$1,699                                |
| Base Cost per Env / yr | \$20,387                               |
| Calculation            | 2 envs × \$1,699/mo = \$3,398/mo total |

|         |          |        |
|---------|----------|--------|
| Monthly | Annual   | 5-Year |
| \$3,398 | \$40,774 | \$0    |

| Category       | Role                                            | Nodes | Instance        | \$/month |
|----------------|-------------------------------------------------|-------|-----------------|----------|
| Kubernetes     | Cloud Managed K8s                               | 1     | EKS             | \$73     |
| Kubernetes     | Worker Nodes (Web, Mobile, WebAPI)              | 2     | m5.4xlarge      | \$1,169  |
| Kubernetes     | Worker Nodes (Graphana & Prometheus) – optional | 1     | m5.2xlarge      | \$321    |
| Kubernetes     | Worker Nodes (EFK) – optional                   | 1     | m5.4xlarge      | \$602    |
| S3             | S3 Storage + Replication                        | 1     | S3              | \$34     |
| Caching        | Caching Service (ElastiCache)                   | 1     | cache.r6g.large | \$121    |
| Infrastructure | Load Balancer (Internal + External)             | 1     | ALB             | \$16     |
| Infrastructure | NAT Gateway                                     | 2     | NAT             | \$32     |
| Infrastructure | Bastion Host                                    | 1     | m5.large        | \$94     |
| Infrastructure | Image Registry (ECR)                            | 1     | ECR             | \$50     |
| Backup         | Back Up – DB & Infra Logs                       | 1     | S3              | \$34     |
| PGSQL DB       | —                                               | —     | db.r5.2xlarge   | \$850    |



# Notes & Assumptions

Detailed pricing basis, scope, and validity

## ■ Pricing Validity — 15 Days Only

This estimate is valid for **15 calendar days** from the date of issue (**09 April 2026**). AWS On-Demand prices, instance availability, and currency exchange rates are subject to change without notice. A revised estimate must be requested if requirements change or the validity period lapses.

## Pricing & Infrastructure Assumptions

|                             |                                                                                                                                                                                                                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS Pricing Basis           | All prices are AWS On-Demand, Linux/UNIX, shared tenancy. No Reserved Instances or Savings Plans applied. Switching to 1-Year Reserved pricing typically reduces compute costs by 30-40%.                                                          |
| Region                      | Estimate uses region: us-east-1. Costs vary by region — a regional multiplier is applied to the base us-east-1 rate.                                                                                                                               |
| Inflation Rate              | 4% per annum applied to all compute and managed-service costs from Year 2 onwards, reflecting typical AWS price adjustments and currency exposure.                                                                                                 |
| High Availability (AZ)      | Production and DR environments are architected as Multi-AZ (High Availability). Pre-Prod/SIT/UAT are Single-AZ to optimize costs while maintaining functional parity.                                                                              |
| Database Licensing          | Postgres is Open-Source (No license cost). SQL Server and Oracle are priced on an AWS RDS 'License Included' basis. Enterprise/BYOL discounts are not applied.                                                                                     |
| Fallback Rates              | If the live AWS Pricing API is unavailable, hardcoded March 2026 on-demand estimates are used. API-sourced prices are marked with a checkmark; estimated prices are noted as estimated.                                                            |
| Node Distribution           | Infrastructure sizing computed by a rule-based algorithm, optionally enhanced by Groq LLM (llama-3.3-70b-versatile) at temperature=0. SHA-256 caching prevents redundant API calls.                                                                |
| Named & Concurrent Users    | Concurrent users are fixed at 30% of Total Named Users for every forecast year. Year-over-Year growth is applied per the rates selected in the estimator tool.                                                                                     |
| Security & Compliance Costs | Fixed INR-converted estimates included: Airtel SOC \$500/mo, SOC Machines Rs.30,000/yr, Req 4 Antivirus Rs.2,600/yr, Req 6 Data Discovery Rs.4,000/yr. These are not subject to AWS price changes.                                                 |
| Pre-Prod / SIT / UAT        | Environments sized at ~40% of Production infrastructure. PostgreSQL uses self-hosted EC2 (Patroni HA). SQL Server / Oracle use AWS RDS Managed. Costs are additive and not included in the Production PUPM.                                        |
| Disaster Recovery (DR)      | DR sized using a Pilot Light / Warm Standby model at ~50% of Production compute. Active-Active DR would approximately double the DR cost shown.                                                                                                    |
| GCP Pricing                 | GCP prices are hardcoded us-central1 On-Demand Compute Engine estimates (March 2026). GCP Committed Use Discounts (1-Year CUD) typically reduce costs by ~37%.                                                                                     |
| PUPM Formula                | $PUPM = \text{Discounted Annual Cost} / 12 / \text{Named Users}$<br>Discounted cost = Total AWS Cost x (1 - 11% discount). Total includes production, DR, Pre-Prod, buffer (5%), Business Support (3.91%), and one-time migration costs in Year 1. |
| Managed Services            | Managed Services fees (30% of AWS annual total) cover platform monitoring, patching, incident management, and lifecycle support by the BusinessNext operations team.                                                                               |

## Scope of This Estimate

|          |                                                                             |
|----------|-----------------------------------------------------------------------------|
| Included | Production cloud infrastructure (compute, database, storage, networking)    |
| Included | Pre-Production / SIT / UAT environments (if selected)                       |
| Included | Disaster Recovery environment (if selected)                                 |
| Included | BusinessNext platform licensing and managed services                        |
| Included | One-time migration, performance testing, and managed-service setup (Year 1) |
| Excluded | Application development or customisation effort                             |
| Excluded | End-user device provisioning or on-site hardware                            |
| Excluded | Third-party SaaS subscriptions (e.g. Salesforce, Twilio, Okta)              |
| Excluded | Data centre co-location or physical network charges                         |
| Excluded | Regulatory compliance certification fees                                    |

Legal Disclaimer

This document is prepared by BusinessNext for the exclusive use of the named client organisation. All cost figures are estimates based on current AWS/GCP published pricing and the inputs provided at the time of generation. Actual costs may vary depending on usage patterns, reserved-instance commitments, enterprise discount programmes, and any changes to cloud provider pricing.

This estimate does **not** constitute a binding commercial offer or contract. A formal Statement of Work (SOW) and Order Form must be executed prior to commencement of any services. BusinessNext reserves the right to revise this estimate upon receipt of updated requirements or upon expiry of the 15-day validity period.

**Confidentiality:** This document is CONFIDENTIAL. It must not be reproduced, distributed, or disclosed to any third party without prior written consent from BusinessNext.